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Transmission-Mounted Power Control Unit with High Power Density for Two-Motor Hybrid System

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Yukiya Kashimura
Honda R&D Co., Ltd.

Yuki Negoro
Keihin Corp

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Abstract

A second-generation power control unit (PCU) for a two-motor hybrid system is proposed. An optimally designed power module, which is a key component of the PCU, is applied to increase heat-resistant temperature, while the basic structure of the first generation is retained and the power semiconductor chip is directly cooled from the single side. In addition to the optimum design, by decreasing the power loss as well as increasing the heat-resistant temperature of the power semiconductors (IGBT: Insulated Gate Bipolar Transistor and FWD: Free Wheeling Diode), the proposed PCU has attained 25% higher power density and 23% smaller size compared to first-generation units, maintaining PCU efficiency (fuel economy). To achieve a high yield rate in the power module assembly process, a new screening technology is adopted at the initial stage of power module manufacturing. In the proposed technology, the maximum current required by hybrid systems can be applied to the component power semiconductor chips at high temperatures before power module assembly. The proposed PCU is directly mounted on the transmission case in the engine compartment of the vehicle and connected to the motor using three-phase connectors. Since conventional three-phase AC cables are eliminated in the layout, space saving and flexibility in the engine compartment layout are provided, simplifying the process of vehicle assembly.

Introduction

Addressing the issues of global warming and air pollution due to CO_2 emissions and energy issues are urgent themes for the automotive industry.

Honda developed a two-motor hybrid system in 2013 as one possible method of addressing these issues (Reference 1). The power train consists of the engine, an e-CVT (electric coupled Continuously Variable Transmission) with built-in motor and generator, a power control unit (PCU) and an intelligent power unit (IPU).

The PCU mounted in this first-generation two-motor hybrid system is located inside the engine compartment to help expand the cabin space. The engine compartment contains large parts such as the motor and generator in addition to the engine, so a large PCU size limits the degree of mounting freedom. In addition, the PCU is located on the body frame directly above the transmission case, and to connect the motor and generator, three-phase AC cables are used. This connection method using cables was also an issue limiting the degree of mounting freedom.

A new PCU with high power performance that can be applied to diverse body frames is needed to promote the spread of electrically driven vehicles such as HEV (Hybrid Electric Vehicle), plug-in HEV (PHEV), EV (Electric Vehicle) and FCV (Fuel Cell Vehicle) in order to address future energy issues. This means that it is necessary to increase the power density and enhance the layout capability of the PCU inside the engine compartment. The authors developed a second-generation PCU for two-motor hybrid system that can be directly mounted on the transmission.

Overall Configuration and Basic Structure

The PCU layout for the second-generation two-motor hybrid system is shown in Fig. 1. The PCU is located inside the engine compartment in the same way as the first-generation system. The PCU is located on the body frame in the first-generation system. The second-generation PCU, on the other hand, is directly mounted on the transmission case (Fig. 2).

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The block diagram of the second-generation PCU is shown in Fig. 3 and its structure is shown in Fig. 4. As well as the first-generation PCU, the second-generation PCU consists of two inverters that drive the motor (which drives the vehicle) and generator (which generates electricity), an electric control unit (ECU) that controls these inverters, and a voltage control unit (VCU) that optimally controls the system voltage. An intelligent power module (IPM) comprises a power module, gate driver and cooling structure. The power semiconductors that are used in the power drive unit (PDU) and VCU are all in one package as a power module case (Fig. 5).

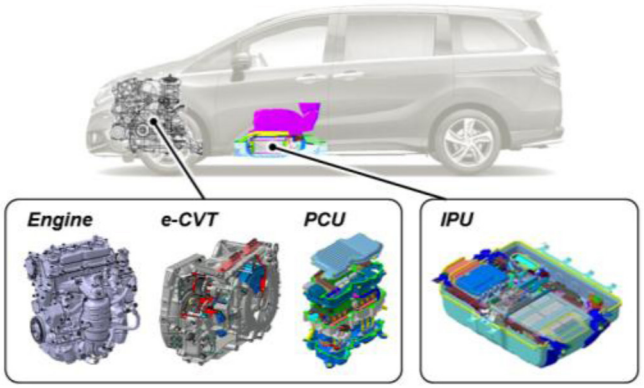


Figure 1. Layout of PCU in two-motor hybrid system

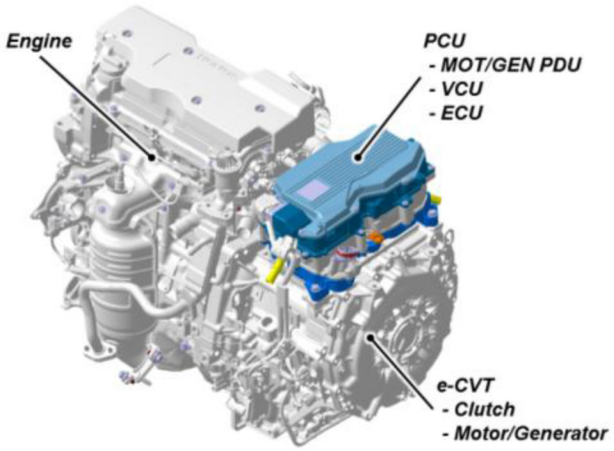


Figure 2. Layout of hybrid power train

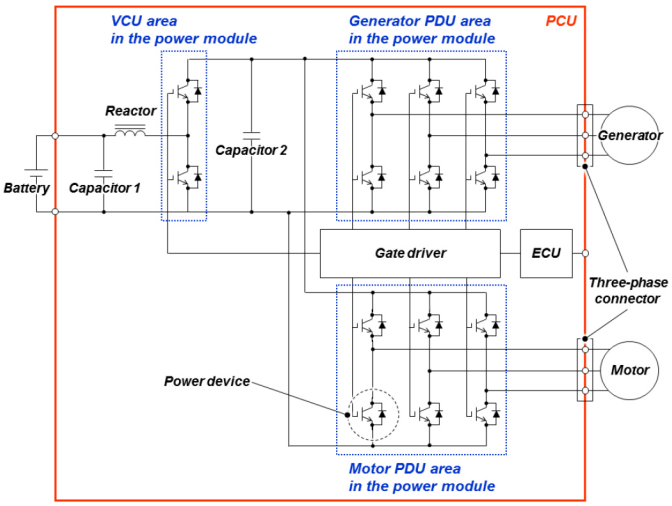


Figure 3. Block diagram of PCU

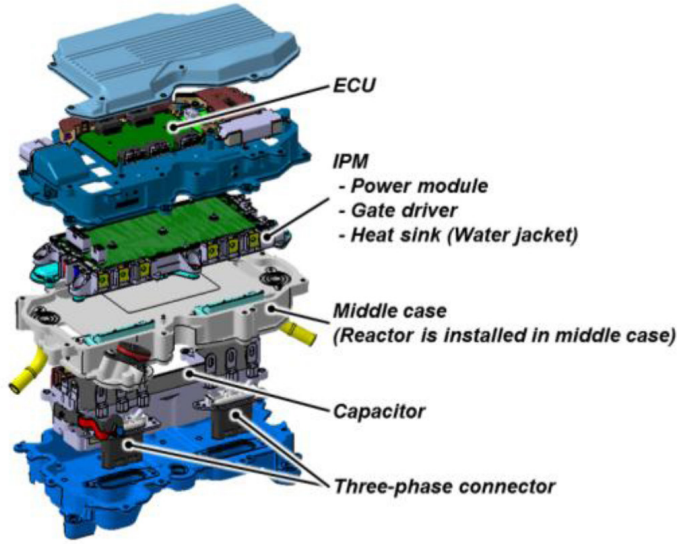


Figure 4. Structure of PCU

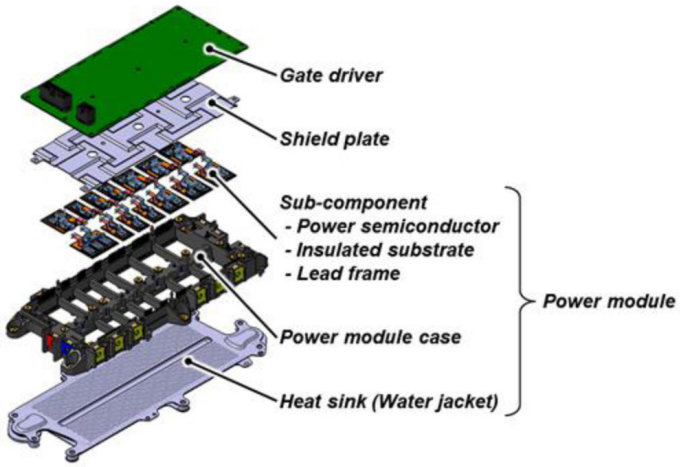


Figure 5. Structure of IPM

HEV System and PCU Specifications

Table 1 shows the specifications of the first- and second-generation HEV system and PCU. For the second-generation system, a high-torque and high-power motor is developed by increasing the output characteristic to meet the driving performance of the vehicle. The volume of the PCU is reduced by 23% with 8.9 L compared to the volume 11.5 L of the first-generation PCU by decreasing the size of the power module, gate driver and VCU through optimization layout of the included components. The power density in relation to volume is increased by 25% with 43.6 kVA/L compared to the power density 34.8 kVA/L of the first-generation PCU (Fig. 6).

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Table 1. Specifications of HEV system and PCU

		First generation	Second generation
HEV system	System maximum voltage [V]	700	700
	Motor maximum torque [Nm]	307	315
	Motor maximum output [kW]	124	135
PCU	Maximum output [kVA]	400	388
	Maximum boost voltage [V]	700	700
	Motor maximum current [Arms]	330	320
	Generator maximum current [Arms]	155	155
	Main components	Motor/Generator inverter Gate driver, ECU VCU	Motor/Generator inverter Gate driver, ECU VCU
	Weight [kg]	17.4	14.3
	Volume [L]	11.5	8.9
	Power density in relation to volume [kVA/L]	34.8	43.6

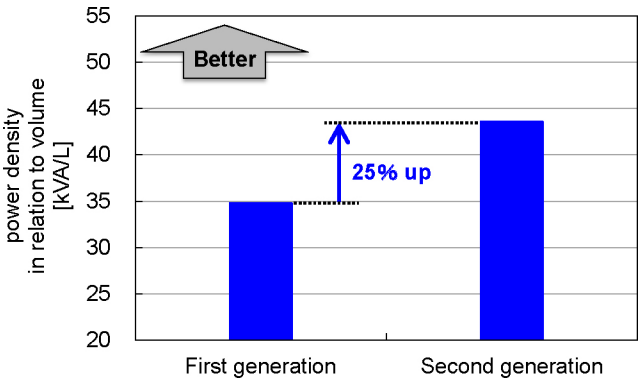


Figure 6. Power density increasing of PCU

Power Module

The size of the power module is relatively large in the PCU as well as the power module being a key component of the PCU. The high power density of the PCU is achieved by reducing the size of the power module while its loss characteristic is maintained. As shown in Fig. 5, the power module mainly consists of power semiconductors and cooling structure. To reduce the size of the power module, we focused on reducing the power loss of the power semiconductors, increasing the heat-resistant temperature and saving space for wiring.

Reducing the Power Loss of the Power Semiconductors

As shown in Figs. 7, 8, both the Insulated Gate Bipolar Transistor (IGBT) and Free Wheeling Diode (FWD) chips used in the second-generation PCU have enhanced performance. Cross-sectional images of the IGBT are shown in Fig. 9. Regarding the IGBTs, the carrier storage structure near the surface of the chip has been modified over the entire chip surface. In the carrier storage region, an electrode does not connect to the semiconductor surface to enhance the carrier storage effect. This increases the conductivity modulation effect during the on-state by decreasing the steady loss (Vce(sat)), and also decreases the dV/dt by modifying the feedback capacitor charge/discharge speed during turn-on states. The voltage surge of the FWDs during recovery states is decreased, so the switching speed in turn-on states is high with low voltage surge. This decreases the turn-on loss (Eon). Regarding the diodes, the dopant profiles of the anode and cathode have been optimized, making it possible to reduce the reverse recovery charge Qrr during reverse recovery and achieve lower loss. The chip size has been reduced by reducing the power loss of the power semiconductors while maintaining the loss characteristic of the power module.

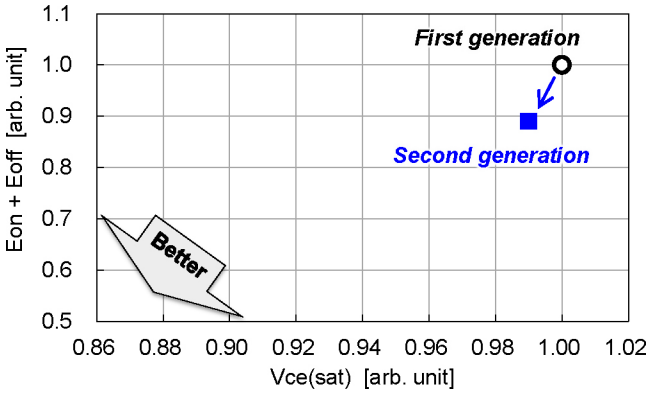


Figure 7. Normalized characteristics of IGBT

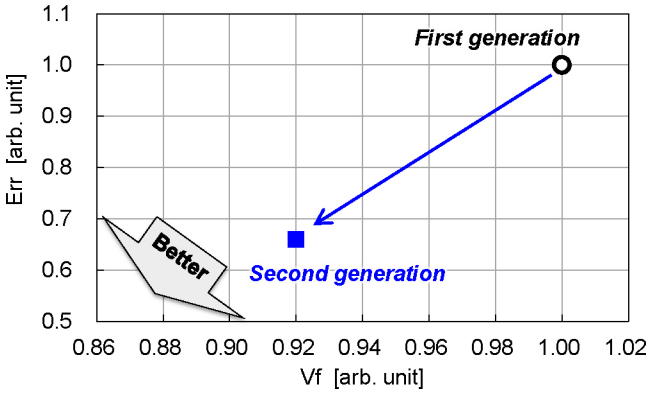


Figure 8. Normalized characteristics of FWD

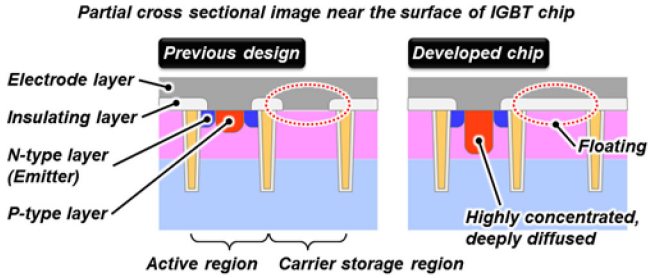


Figure 9. Structure of IGBT

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Increasing the Heat-Resistant Temperature of the Power Semiconductors

Reduction of the pinch-resistance inside the IGBT with a highly concentrated, deeply diffused p-type semiconductor enabled expansion of the safe operating area (SOA) (Fig. 9). This increases the maximum allowable junction temperature during operation to 175°C. Measures were taken at the same time so that all reliability items required for the chip were assured at 175°C. This allowed increasing the allowable power dissipation, which contributed to reducing the chip size. As a result, the IPM and PCU were designed on the premise that the maximum heat generation temperature of the chips is 175°C.

Screening Technology for Power Semiconductors

In power modules such as the first- and second-generation that contain multiple power chips, a single defective power chip renders the entire module defective. Therefore, screening inspection was performed for the first-generation PCU in the sub-assy state with multiple power chips mounted before assembly into the power module. In addition, inspection is performed by flowing current in the sub-assy state, so a land for electric contact is needed on the DCB (Direct Copper Bond) insulated substrate, and wiring is also needed for relays. This resulted in a large sub-assy substrate size.

A typical issue for power chip screening inspection thus far is that go/no-go screening could not be performed adequately in the chip state, so go/no-go screening was performed in the screening inspection after sub-assy assembly in a later process. The reason for this is that the contactor of the test machine that contacts the power chip in the power chip sorting inspection, which should be performed in the element state, has a small current capacity.

For the second generation, an inspection system capable of applying current of 1800 A/cm^2 or more in the chip state was constructed by developing an original exchangeable contactor with narrow-pitch multiple contact points (Fig. 10) (Reference 2). The image of current-voltage locus during go/no-go screening inspection is shown in Fig. 11. The current-voltage locus, that is higher than the current-voltage locus during vehicle operation, can be applied to the chip during screening inspection. This greatly increased the defect identification rate in the chip state, making it possible to achieve a high yield rate for the sub-assy and power module. In addition, this technology also contributed to sub-assy substrate size reduction, resulting in power module size reduction.

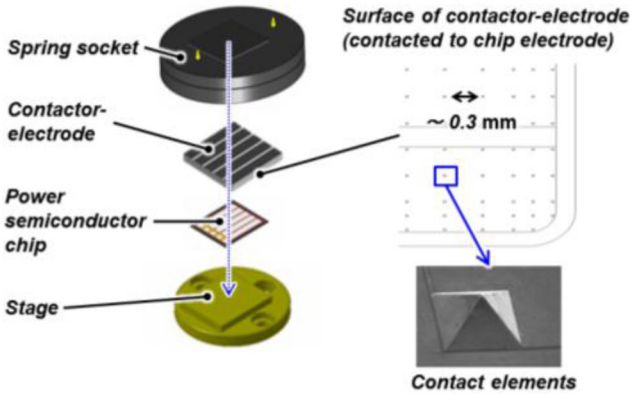


Figure 10. Exchangeable contactor with narrow-pitch multiple contact points

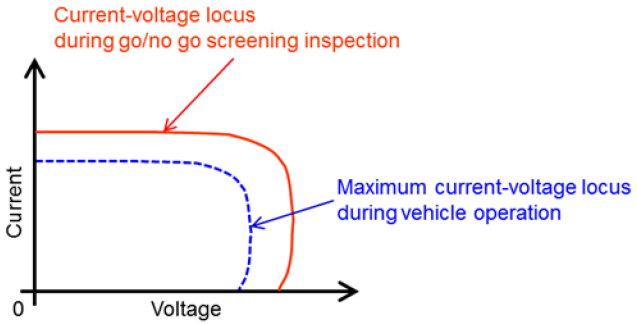


Figure 11. Image of current-voltage locus during go/no go screening inspection

Power Block Wiring Design

In the power module of the first-generation PCU, the I/O bus bar and power semiconductors were ultrasonically joined using aluminum wires. A single power module requires 1,000 or more aluminum wires, so multiple facilities were required and manufacture took much time. In addition, the space to bond the aluminum wire to the I/O bus bar is required, preventing reduction of the size of the power module. Therefore, in the second generation the aluminum wire was replaced with a lead frame that is a copper foil pressed part, and a new wiring method was adopted that connects the I/O bus bar and power semiconductors with solder (Fig. 12). Increasing the heat-resistant temperature is necessary, as the maximum heat-generation temperature of the power semiconductors increases. Optimization of the lead frame thickness reduced the thermal stress between the power semiconductors and the lead frame, and secured reliability. This optimization reduced the power module size through space saving in the wiring structure and shortened the manufacturing process.

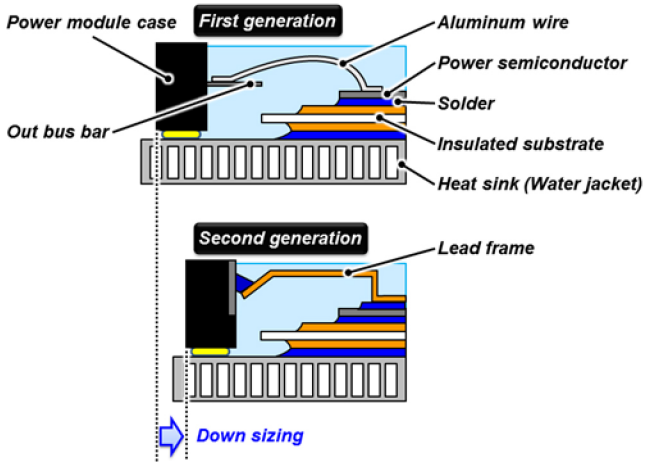


Figure 12. Wiring design

Cooling Structure

The second-generation power module retains the cooling structure of the first generation, and has a basic structure that directly cools the power semiconductor chip with water from one side. Enhancement of radiation performance and reduction of the power semiconductor chip size are effective means to reduce the power module size. The authors therefore focused on reduction of thermal resistance in order to further enhance radiation performance.

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Considering the heat sink plate thickness and the solder fatigue strength, a second-generation cooling structure with high heat dissipation and high heat resistance was devised. A closed water channel-type heat sink structure was used to secure high bending hardness even if the heat sink plate thickness is reduced in order to decrease the heat resistance. The solder fatigue strength is increased by adopting a high-ductility solder. By the optimization of the heat sink aluminum plate thickness and the DCB insulated substrate thickness, heat resistance of the power module has been decreased.

This power module with integrated closed water channel-type heat sink achieved a reduction in thermal resistance of 29% compared to the first generation, and made it possible to reduce the power semiconductor chip size. In addition, reduction of the aluminum heat sink weight also realized a reduction in the PCU weight.

Gate Driver and Electronic Control Unit

In order to decrease the size of the IPM, it is also necessary to downsize the printed circuit board (PCB) of the gate driver and electronic control unit (ECU). To decrease the number of component, circuit component integration was attempted in the gate driver PCB. Comparison between the gate drive PCB of the first and second generations is illustrated in Fig. 13. In the first generation, the gate drive circuit, voltage source control circuit, temperature monitoring circuit and isolation devices (optocouplers) were separately composed. In the second generation, these circuits, which were composed of discrete components, were integrated in a single package. In addition, magnetic coupling technique was adopted in isolation devices. Although they are resistant in environments of high external magnetic flux density, the layout of the isolation devices at the PCB was designed to avoid areas of high magnetic flux. Through this integration, 31% of the number of components and 21% of the size were reduced at the gate driver PCB.

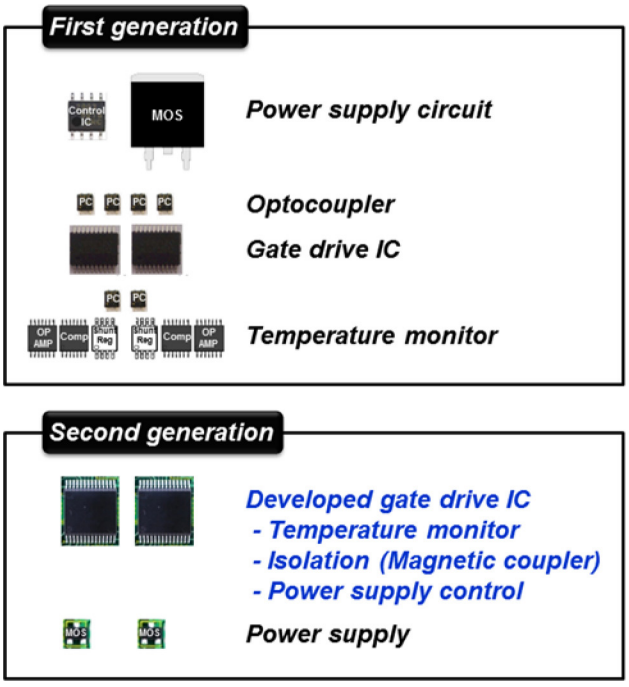


Figure 13. Gate drive IC

To decrease the number of components, circuit integration was also attempted in the ECU. Comparison between the microprocessor block adopted in the first and second generations is illustrated in Fig. 14. In the first generation, three CPU chips were utilized in order to respectively control the motor, generator and VCU. In the second generation, not only these three CPU chips were integrated in a single package, but also the RDC (Resolver to Digital Converter) was included.

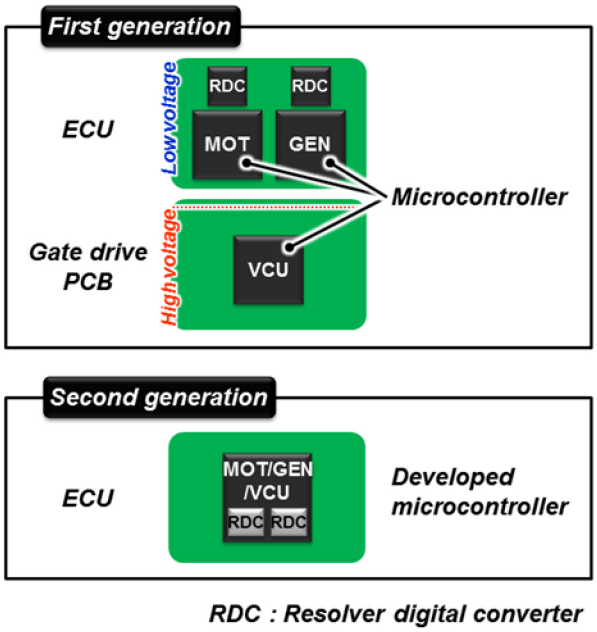


Figure 14. Microcontroller

Voltage Control Unit (VCU)

Since the reactor and capacitors of the VCU occupy a high volume rate in the PCU, downsizing these components leads to downsizing the PCU. By decreasing power loss and increasing the heat resistance of the power semiconductor chips, the size of power module and reactor could be decreased in well balance. As a result, the VCU carrier frequency increased 1.6 times compared to the first generation. Conventionally, to increase the carrier frequency, increasing the size of the power semiconductor chips is required. However, this was suppressed since decreasing power loss and increasing the heat resistance were realized. As a result, the inductance value necessary for the VCU to operate in the boost mode is reduced, and the size of the reactor could be reduced by 26%.

The structure of the reactor was also modified in order to improve noise vibration performance (Fig. 15). While a rigid structure was adopted in the first generation, a floating structure was implemented in the second generation. By isolating the electromagnetic part from the case with stays and resin material, the noise vibration transfer could be suppressed. In relation to the capacitor, the adoption of a thin-film material and decrease of components achieved a 27% size reduction and 23% weight reduction compared to the first generation.

These achievements contributed to the high power density of the PCU.

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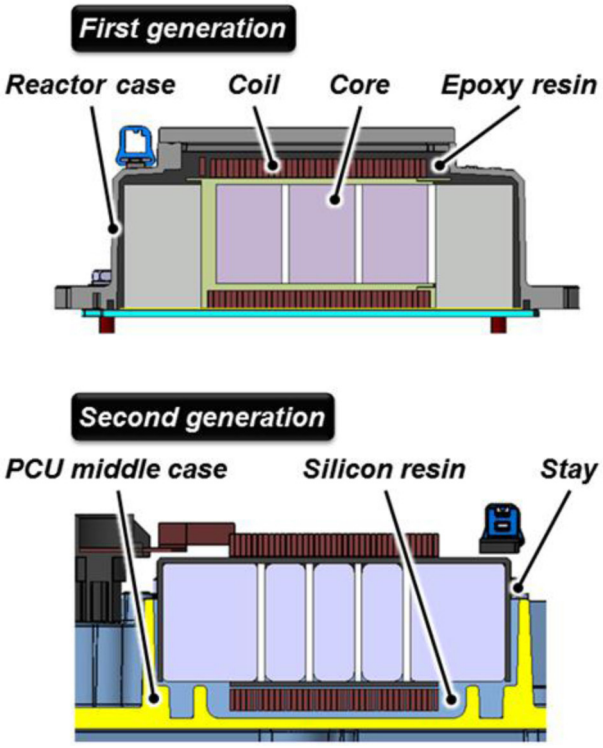


Figure 15. Structure of reactor

Transmission-Mounted Design

The mounted design of PCU in the first- and second- generations is shown in Fig. 16. The first-generation PCU was positioned on the body frame above the transmission case, and the wiring with the motor and generator used three-phase AC cables. Large current flowed in the three-phase AC cables, so a large-diameter wire was required. For this reason, the cable bending curvature was limited, which restricted the degree of freedom of the mounting layout. Therefore, the second-generation PCU is directly mounted on the transmission case and has a structure that connects to the motor using three-phase connectors.

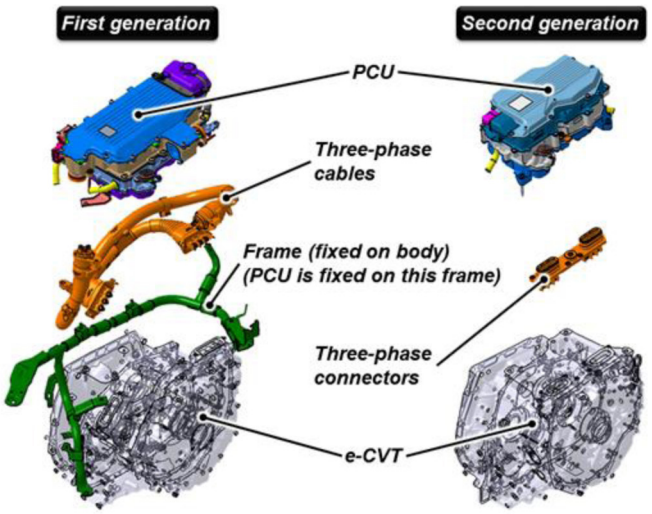


Figure 16. Layout of PCU

Securing reliability in relation to vibration and heat is an issue when mounting the PCU directly on the transmission. In addition to the vibration from the road surface, the PCU also needs to be able to endure vibration from the power plant, such as engine combustion and rotation, and motor rotation. In addition, vibration from other external factors is amplified within the PCU, so the PCU internal components require a vibration-resistant design. Furthermore, the use of a directly mounted structure increases the heat transferred to the PCU from the motor, so it is necessary to secure adequate heat resistance.

Regarding vibration resistance, reducing the size of PCU internal components mainly on the PCB is important. A transformer and electrolytic capacitor are relatively large parts in the gate driver PCB and ECU. A PCU internal component design for vibration is shown in Fig. 17. On the gate driver PCB, the produced stress during vibration is reduced by distributing the transformer, reducing its size and adopting ceramic capacitor. On the ECU, the stress is reduced by size reduction of the electrolytic capacitor. In addition, the fastening points to the power module are optimized to suppress sympathetic vibrations (Fig. 18).

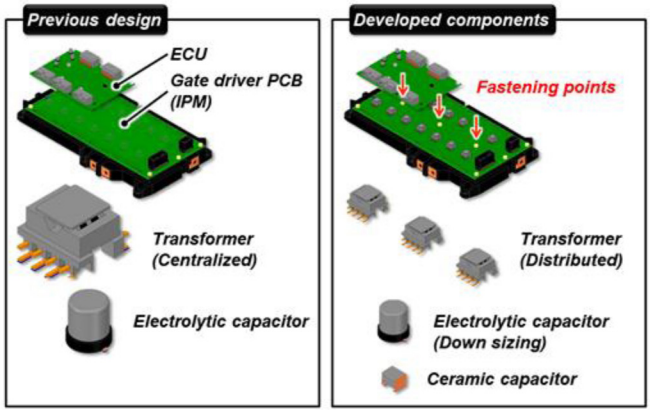


Figure 17. PCU internal components design for vibration

Regarding heat resistance, the PCU internal components were designed so that the temperature does not exceed the maximum temperature of the heat transferred to the PCU from the motor.

The use of a transmission-mounted design that secures both vibration and heat resistance as described above and the elimination of the three-phase AC cables achieved a space-saving layout inside the engine compartment and also enabled simplification of the assembly process. In addition, the use of three-phase connectors simplified the three-phase bus bar inside the PCU and also contributed to reducing the PCU size.

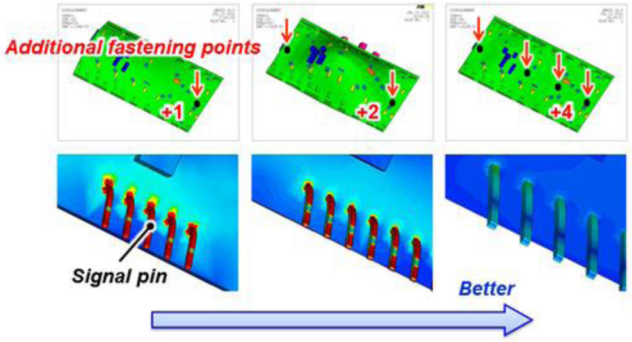


Figure 18. Vibration simulation for gate driver PCB

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Summary

A second-generation PCU for a two-motor hybrid system was developed by developing the following technologies with the aim of increasing the power density and enhancing the vehicle layout capability.

- Reduction of chip size by reducing the chip loss and increasing the chip heat resistance
- Reduction of chip size by increasing the heat resistance of the power module and enhancing radiation performance
- Reduction of the gate driver substrate size by using a gate driver IC with built-in insulation and power control
- Reduction of the ECU substrate size by using only one microcontroller
- Reduction of the VCU size by increasing the carrier frequency
- Achievement of a space-saving layout inside the engine compartment through a transmission-mounted structure and connection using three-phase connectors

The above enhancements achieved a PCU with power of 388 kVA and a volume of 8.9 L, and increased the power density by 25%. These technologies enhance the vehicle layout capability and can be applied not only to HEV but also to PHEV, EV and FCV.

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